LuoLinguaAI: Phase One Progress Report

1. Purpose of this report

This report explains what has been built for LuoLinguaAI, what still needs to be done, and what belongs to Phase Two. The PRD describes a large system with 15 main modules. The project discussion set a six-week period for Phase One. The one-week period was for an early version that can be seen and reviewed. It was not the full six-week Phase One.

The report uses the PRD, the project files, and the current system so you can open the deployed system and review the work.

2. Overall progress

The first working version of the web system has been built and deployed. The database is running. Users can open the system on a phone or computer, and some Luo artifact pictures have been added and can be seen by users.

This is real progress toward Phase One. It does not mean that every part of the PRD is complete. The main work still needed is to add and check more approved Luo content, complete the agreed Phase One features, and test the main tasks.

3. Phase One achievements

Area	What has been achieved
Web system	The web system has been built and deployed using Vercel. It can be opened through a web link.
Phone and computer access	The system can be installed from the browser. It was tested on a phone and on a laptop, and its app icon appears on both screens, making it easy to open like an app.
Database	The Neon database is running and connected to the system, storing system information such as users and cultural records.
System design	The database has places for language settings, user roles, cultural information, dictionary entries, translations, pictures and other media, transcripts, and AI search information.

Area	What has been achieved (Cont.)
Public pages	The system has pages for key content areas, including the dictionary, proverbs, riddles, oral histories, stories, songs, artifacts, and heritage sites.
Admin tools	Admin pages and tools have been built for managing users and content, reviewing submissions, managing media, and viewing system information.
User access	Login and account features, including admin roles, have been built into the system. The public Google sign-in has been configured for the current version.
Artifact pictures	Some artifact pictures have been uploaded. Users can see them in the system.
AI search and chatbot	Connected to Gemini with code for semantic search and RAG chatbot (retrieving stored records to assist answer generation).
Transcript tools	Code exists to make a short summary of a transcript and identify names of people, places, and clans (pending checks against approved Luo source material).

> **Phone installation: what it means:** The phone and laptop installation is an installable web app: the browser puts an icon on the screen and opens the website in an app-like window. It is not yet a separate Android or iOS app published through Google Play or Apple's App Store.

4. Phase One work still to complete

Work still needed	Why it matters
Add more approved Luo content	Source archive has many transcripts, reports, and pictures, but they have not all been added to the live system so there is enough real content to review main sections.
Finish selected Phase One sections	Pages exist for several modules, but each needs enough checked content and a complete user experience.
Complete virtual Luo museum	Artifact pictures are visible, which is a start. More approved artifact and heritage-site records, descriptions, and pictures are needed. A full 3D or virtual-tour experience is a larger feature and needs a separate decision.

Work still needed (Cont.)	Why it matters
Add and check available media	Current source files include pictures, but no audio or video files were found. Audio and video features need the project team to provide files and confirm sharing rights.
Test the main tasks	Check viewing/search, admin additions/approvals, phone/computer image uploads, and live AI search/chatbot using real approved records.
Check AI answers	Chatbot should give answers based on approved Luo content and show which records it used. People who know the culture should review answers for accuracy.
Add rest of approved data	Prepared transcript data needs to be carefully moved into the system with source/access info. Having a file in the folder doesn't automatically put it in the live DB.
Review access and consent	Some transcripts are marked research or internal use. Before public display or external AI use, the team must confirm this use is allowed.
Complete final handover	Prepare simple instructions for using the system, managing content, backing up the database, and reporting problems.

Note: The supplied NLP package describes 26 prepared transcripts divided into 2,708 text sections. This is useful source material. It is not proof that all these sections are already in the live system or approved for public use.

5. Phase One modules from the PRD

PRD area	Current status
Dholuo dictionary & Proverbs	Pages and database support present. Small sample available; more checked words, proverbs, meanings, and examples needed.
Riddles & Oral histories	Quiz feature, pages, and transcript support present. Small sample available; more approved transcripts and explanations needed.
Folktales, Songs & Artifacts	Pages present. Collection details, lyrics, approved sound recordings, and records need expansion. Artifact pages have visible pictures.
Heritage sites, Chatbot & Others	Map components, chatbot/RAG code present. Tutor, learning centre, calendar, AV library, and research portal are incomplete or stubbed.

6. Phase Two: later work

Phase Two should begin after agreed Phase One work is reviewed. Uncompleted PRD features: separate Android/iOS apps, offline mode with auto-sync, speech-to-text / text-to-speech, pronunciation practice, full AI language tutor, OCR, machine translation, image recognition, cultural knowledge graph, full lessons/progress, 3D artifact models, story animation, virtual tours, research portal, AV library, full multi-language support (Kikuyu is inactive), and university data-centre migration.

Phase Two groundwork already present: Phone-install feature, early transcript summary and name-finding tools, and database starter entries for future learning, games, research, and additional languages.

7. Important note about AI

Current AI features use Gemini to work with system information. The system has not trained a new Luo language model; text vectors help find related information. The AI should only use approved project information, and its answers and summaries should be reviewed by culture and language experts.

8. Recommended next steps for Phase One

- Confirm which content sections must be ready for the Phase One review.
- Confirm which transcripts, pictures, and other media can be shown publicly and used with AI.
- Add a useful set of approved content to selected sections, including the artifact collection.
- Test main public, admin, media, and AI tasks on the deployed site using phone and computer.
- Record feedback and agree which remaining tasks belong to Phase One vs. Phase Two.

9. Closing statement

Phase One has made clear progress: the web system is deployed, the Neon database is running, the system can be installed from phone/laptop browsers, admin and content features are built, artifact pictures are visible, and initial AI tools are in place.

The main unfinished work is adding approved content and media, completing/testing selected Phase One features, and reviewing the system. Broader mobile apps and advanced AI features belong to Phase Two unless the project team formally changes the agreed scope.